FLUID MECHANICS

Subject Code: 23A01303

UNIT - 4: Fluid Dynamics

B.Tech / JNTUA R23 / Handwritten Notes

Syllabus: Surface and body forces; equations of motion – Euler's equation; Bernoulli's equation – derivation; energy principle; practical applications of Bernoulli's equation: venturimeter, orifice meter and Pitot tube; momentum principle; forces exerted by fluid flow on pipe bend; vortex flow – free and forced; definitions of Reynolds number, Froude number, Mach number, Weber number and Euler number.

1. Forces Acting on a Fluid in Motion

Newton's second law applied to a fluid particle: $F = m\,a$. The forces acting are classified as:

Type	Meaning	Examples
Surface forces	Forces acting on the boundary (surface) of a fluid element by the surroundings; proportional to area	Pressure force F_p, viscous (shear) force F_v, surface tension F_s
Body forces	Forces acting on every mass element throughout the volume; proportional to mass/volume	Gravity $F_g = mg$, magnetic and electric forces, centrifugal force
Inertia force	Rate of change of momentum of the particle (mass × acceleration)	$F_i = m\,a$

- Forces influencing fluid motion in practice: gravity, pressure, viscosity, turbulence, surface tension and compressibility.
- Net force $F_x = (F_g)_x + (F_p)_x + (F_v)_x + (F_t)_x + (F_s)_x + (F_c)_x = m\,a_x$ – Reynolds equation of motion (all forces).
- Neglecting turbulence, surface tension and compressibility: $F_g + F_p + F_v = m\,a \rightarrow$ Navier–Stokes equation.
- Neglecting also viscosity (ideal fluid): $F_g + F_p = m\,a \rightarrow$ Euler's equation of motion.

2. Euler's Equation of Motion

Consider a small cylindrical element of length ds and cross-section dA along a streamline in steady flow of an ideal fluid. Let θ be the inclination of the streamline with the horizontal.

Fluid Dynamics

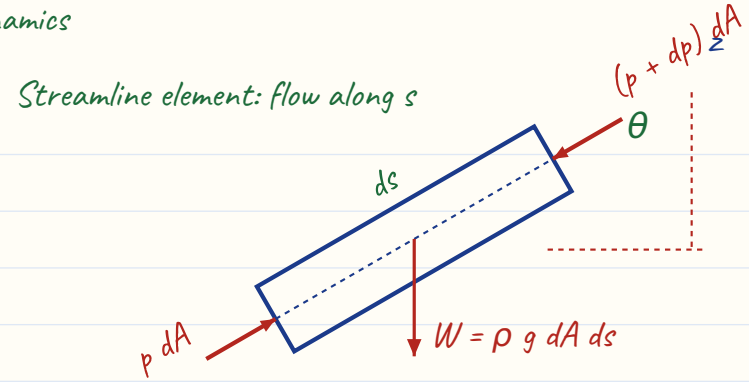


Fig 4.1: Forces on a fluid element along a streamline

- Pressure force on the upstream face = $p\,dA$; on the downstream face = $(p + dp)\,dA$ (opposing).
- Weight component along the streamline = $\rho\,g\,dA\,ds\cos\theta = \rho\,g\,dA\,dz$ (since $ds\cos\theta = dz$).
- Mass of the element = $\rho\,dA\,ds$; acceleration along s: $a_s = V\,dV/ds$ (steady flow).

Newton's second law along s: $p\,dA - (p + dp)\,dA - \rho\,g\,dA\,ds\cos\theta = \rho\,dA\,ds \cdot V\,dV/ds$

$$-dp\,dA - \rho\,g\,dA\,dz = \rho\,dA\,V\,dV$$
$$dp/\rho + g\,dz + V\,dV = 0 \text{ (Euler's equation)}$$

- Valid for steady flow along a streamline of an ideal (inviscid) fluid; it holds for compressible as well as incompressible fluid (ρ may vary with p).
- In 3-D form (Cartesian): $-(1/\rho)\,\partial p/\partial x + X = du/dt$, etc., where X, Y, Z are body force components per unit mass.

3. Bernoulli's Equation

For incompressible flow (ρ = constant), integrate Euler's equation along the streamline:

$$\int dp/\rho + g\int dz + \int V\,dV = \text{const} \rightarrow p/\rho + g\,z + V^2/2 = \text{const}$$
$$\text{Dividing by } g\text{: } p/(\rho\,g) + V^2/(2g) + z = \text{constant} = H$$

Bernoulli's Theorem
For a steady, irrotational (frictionless), incompressible flow of an ideal fluid, the total energy per unit weight (sum of pressure head, velocity head and datum head) remains constant along a streamline.

Term	Name	Meaning	Unit
$p/(\rho g)$	Pressure head	Flow work (pressure energy) per unit weight	m
$V^2/(2g)$	Velocity (kinetic) head	Kinetic energy per unit weight	m

z	Datum (potential) head	Potential energy per unit weight	m
H	Total head	Total energy per unit weight	m

- Between two sections 1 and 2: $p_1/\rho g + V_1^2/2g + z_1 = p_2/\rho g + V_2^2/2g + z_2$
- In terms of pressure: $p + \tfrac{1}{2}\rho V^2 + \rho g z = \text{const}$ (static + dynamic + hydrostatic pressure).
- In terms of energy per unit mass: $p/\rho + V^2/2 + g z = \text{const}$ (J/kg).

Assumptions in Bernoulli's Equation

1. Fluid is ideal (no viscosity, no shear stress, no friction loss).
2. Flow is steady.
3. Flow is incompressible (ρ = const).
4. Flow is along a streamline (for irrotational flow, the constant is the same throughout the field).
5. Only gravity and pressure forces act (no heat transfer, no shaft work, no turbulence or surface tension).
6. Flow is continuous with no addition or removal of energy between the sections.

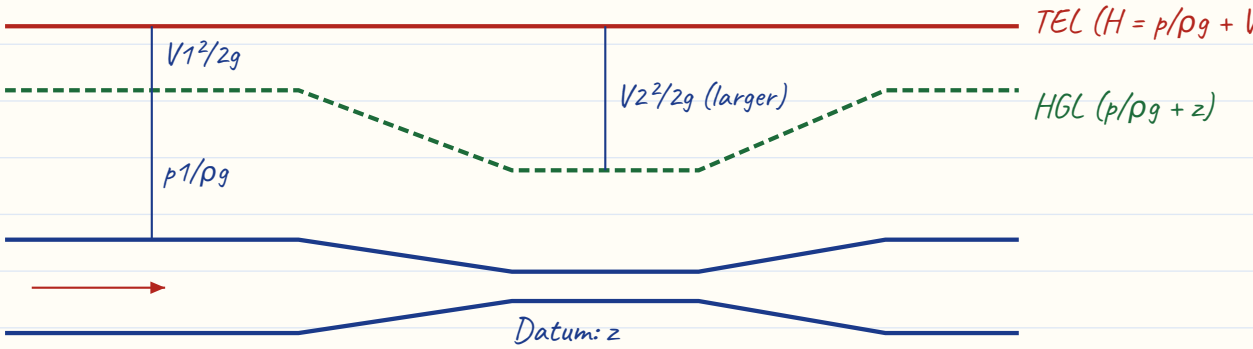


Fig 4.2: Bernoulli's equation – total energy line is horizontal for an ideal fluid

Modified Bernoulli Equation – Energy Principle for Real Fluids

For real fluids, some energy is lost due to friction (h_ℓ); a pump adds head (h_p) and a turbine extracts head (h_t). The first law of thermodynamics for steady flow gives the energy equation:

$$p_1/\rho g + \alpha_1 V_1^2/2g + z_1 + h_p = p_2/\rho g + \alpha_2 V_2^2/2g + z_2 + h_\ell + h_t$$

- α = kinetic energy correction factor (= 1 for uniform velocity, 2 for laminar pipe flow, ≈ 1.05–1.1 for turbulent flow; usually taken as 1).
- Power of a pump or turbine: $P = \rho g Q H$ (watt); shaft power = P/η.
- Bernoulli's equation is a statement of conservation of mechanical energy; the energy equation extends it to include losses and machines.

Example 1 – Bernoulli's equation

Water flows at 0.04 m^3/s through a pipe that reduces from 200 mm (section 1, gauge pressure 50 kPa, elevation 0) to 100 mm (section 2, elevation 3 m above section 1). Neglecting losses find p_2.

Sol: $V_1 = Q/A_1 = 0.04/0.03142 = 1.273$ m/s; $V_2 = 0.04/0.007854 = 5.093$ m/s

$p_1/\rho g = 50000/9810 = 5.097$ m; $V_1^2/2g = 0.0826$ m; $H = 5.097 + 0.083 + 0 = 5.179$ m

$p_2/\rho g = H - V_2^2/2g - z_2 = 5.179 - 1.322 - 3 = 0.857$ m → $p_2 = 0.857 \times 9810 = 8.41$ kPa (gauge)

Example 2 – Pump power (energy equation)

A pump lifts 0.05 m^3/s of water from a sump to a tank 20 m above; total head loss in pipes is 3 m. Find the pump head and the shaft power if $\eta = 70\%$.

Sol: Head to be supplied $H = 20 + 3 = 23$ m. Water power $= \rho g Q H = 9810 \times 0.05 \times 23 = 11281\ W \approx 11.28$ kW

Shaft power $= 11.28/0.70 = 16.1$ kW

4. Practical Applications of Bernoulli's Equation

(a) Venturimeter

A device used to measure the discharge in a pipe. It consists of (i) a converging cone (angle ≈ 20°), (ii) a short cylindrical throat, and (iii) a gradually diverging cone (angle 5°–7°) which recovers pressure with low loss.

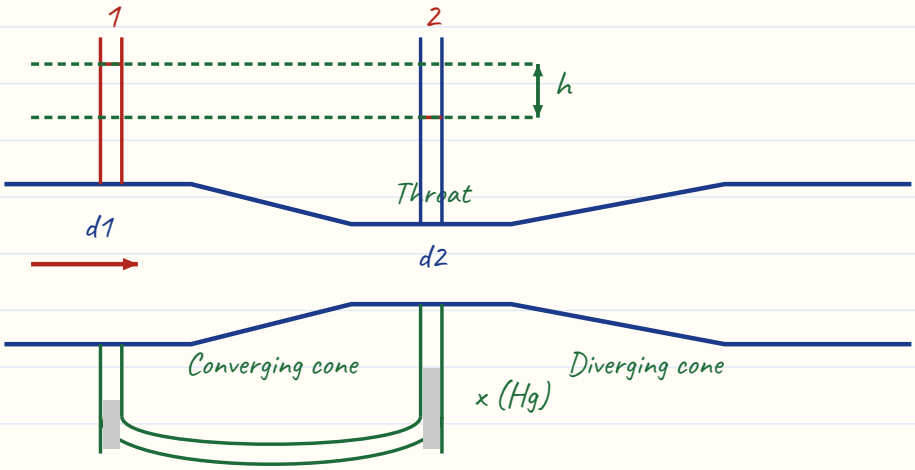


Fig 4.3: Venturimeter with differential manometer

Apply Bernoulli between section 1 (inlet, area a_1) and section 2 (throat, area a_2) for a horizontal meter ($z_1 = z_2$), ideal flow:

$$p_1/\rho g + V_1^2/2g = p_2/\rho g + V_2^2/2g \rightarrow h = (p_1 - p_2)/\rho g = (V_2^2 - V_1^2)/2g$$

From continuity $a_1 V_1 = a_2 V_2 \rightarrow V_1 = a_2 V_2/a_1$

$$Q_{th} = a_1 a_2 \sqrt{(2gh)} / \sqrt{(a_1^2 - a_2^2)}$$

$$Q_{act} = C_d \cdot a_1 a_2 \sqrt{(2gh)} / \sqrt{(a_1^2 - a_2^2)}$$

- h = differential pressure head (m of the flowing liquid) = $x (S_m/S_f - 1)$ for a U-tube differential manometer with manometric liquid of S_m, x = reading. For mercury and water: $h = 12.6\,x$.
- For a manometer with a lighter liquid (inverted U-tube): $h = x (1 - S/S_f)$.
- C_d = coefficient of discharge (0.96 – 0.98) accounts for losses; always < 1.
- Inclined / vertical venturimeter: the same formula holds, but $h = (p_1/\rho g + z_1) - (p_2/\rho g + z_2)$ – the piezometric head difference, read directly from the differential manometer.
- Merits: accurate, low head loss, self-cleaning; Demerits: costly, long, not flexible.

Example 3 – Venturimeter

A horizontal venturimeter has inlet 300 mm and throat 150 mm diameter. A mercury differential manometer reads 0.2 m. Find the discharge for $C_d = 0.98$.

Sol: $h = x (S_m/S - 1) = 0.2 \times 12.6 = 2.52$ m of water. $a_1 = 0.07069\ m^2$; $a_2 = 0.01767\ m^2$

$\sqrt{(a_1^2 - a_2^2)} = \sqrt{(0.0049966 - 0.0003123)} = 0.06844$; $\sqrt{(2gh)} = \sqrt{(2 \times 9.81 \times 2.52)} = 7.031$

$Q = 0.98 \times (0.07069 \times 0.01767) \times 7.031/0.06844 = 0.1258\ m^3/s \approx 125.8\ L/s$

(b) Orifice Meter

A thin plate with a sharp-edged circular hole (diameter $d_0 \approx 0.5\, d_1$) fixed concentrically in the pipe between flanges. As the fluid passes, the jet contracts to a minimum area called the vena contracta (section 2, $a_c = C_c\, a_0$), where the velocity is maximum and pressure minimum. Pressure difference between a section upstream (1) and the vena contracta (2) is measured by a differential manometer.

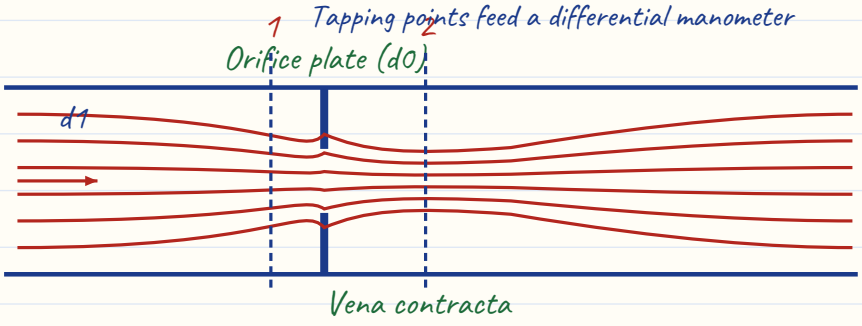


Fig 4.4: Orifice meter

$$Q = C_d \cdot a_0 a_1 \sqrt{(2gh)} / \sqrt{(a_1^2 - a_0^2)}$$

- Derived like the venturimeter (Bernoulli + continuity) with C_d including the contraction coefficient ($C_d \approx 0.6 - 0.65$, much lower than for a venturimeter).
- Merits: cheap, compact, easy to install. Demerits: large permanent pressure (head) loss (since jet expansion causes eddies), lower accuracy, clogging by suspended particles.

Example 4 – Orifice meter

A 200 mm pipe has a 100 mm sharp-edged orifice ($C_d = 0.6$). A mercury manometer reads 0.15 m. Find the discharge of water.

Sol: $h = 0.15 \times 12.6 = 1.89$ m; $a_1 = 0.031416$, $a_0 = 0.007854$ m^2; $\sqrt{(a_1^2 - a_0^2)} = 0.030418$; $\sqrt{(2gh)} = 6.0895$

$Q = 0.6 \times (0.031416 \times 0.007854) \times 6.0895/0.030418 = 0.0296$ m^3/s ≈ 29.6 L/s

Feature	Venturimeter	Orifice meter
Coefficient of discharge	0.96 – 0.98	0.60 – 0.65
Head loss	Small (gradual diverging cone)	Large (sudden expansion)
Cost / size	Costly, long	Cheap, compact
Accuracy	High	Moderate

(c) Pitot Tube

A device to measure the local velocity at a point in a flow. A glass tube bent at 90° is placed with its open end facing the flow. The fluid at the open end is brought to rest (stagnation point) and rises in the vertical limb by height h above the free surface (or static pressure level).

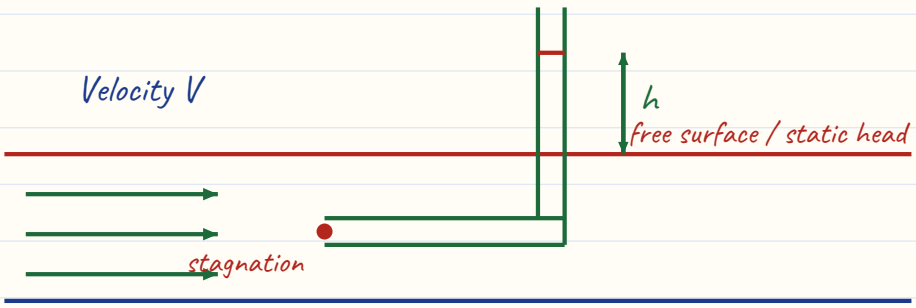


Fig 4.5: Pitot tube in an open channel

Apply Bernoulli between a point 1 upstream (velocity V, static head $p/\rho g$) and the stagnation point 2 ($V_2 = 0$):

$$p/\rho g + V^2/2g = p_0/\rho g \rightarrow h = V^2/2g$$
$$V = \sqrt{(2 g h)} \text{ and actual } V = C_v \sqrt{(2 g h)}$$

- p_0 = stagnation pressure = static pressure + dynamic pressure ($\frac{1}{2}\rho V^2$). C_v = coefficient of the pitot tube ≈ 0.98–1.0.
- Pitot-static tube: a concentric tube arrangement, the inner tube senses stagnation pressure and the outer one with side holes senses static pressure. Differential manometer reads h: $V = C_v \sqrt{(2 g h)}$ with h in metres of flowing fluid.
- For gas flow with a liquid manometer: h = (ρ_manometric/ρ_gas) × x. Used in aircraft airspeed indicators and ducts.

Example 5 – Pitot-static tube in air duct

A pitot-static tube in an air duct (ρ = 1.2 kg/m^3) shows a water-manometer deflection of 25 mm. Find the air velocity (C_v = 1).

Sol: h_{air} = (1000/1.2) × 0.025 = 20.83 m of air; $V = \sqrt{(2 \times 9.81 \times 20.83)}$ = 20.2 m/s

5. Momentum Principle

Impulse–Momentum Equation

The net external force acting on a control volume of fluid in a given direction equals the rate of change of momentum of the fluid in that direction (outflow of momentum minus inflow of momentum).

$$\Sigma F = d(mV)/dt = \rho Q (V_2 - V_1) \text{ (steady flow, one inlet/outlet)}$$
$$\Sigma F_x = \rho Q (V_{2x} - V_{1x}), \Sigma F_y = \rho Q (V_{2y} - V_{1y})$$

- ΣF includes pressure forces on the control surfaces, body forces (weight) and the force exerted by the solid boundary on the fluid.
- The force exerted by the fluid on the solid is equal and opposite to that exerted by the solid on the fluid.
- Momentum correction factor $\beta = (1/A V^2) \int v^2 dA$: = 1 for uniform flow; 1.33 for laminar pipe flow; 1.02–1.05 for turbulent flow.
- Used when the force is required but the energy losses in the flow are unknown (pipe bends, nozzles, jets striking plates, hydraulic jump, turbines).

Force Exerted by Flowing Fluid on a Pipe Bend

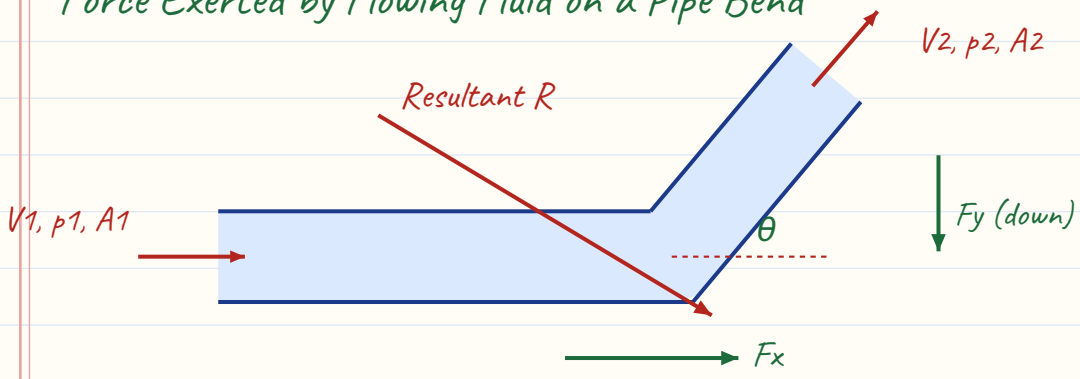


Fig 4.6: Forces on a horizontal pipe bend (deflection)

A pipe bend turns the flow through angle θ, the section changes from A_1 to A_2. Let p_1, p_2 be the gauge pressures. Take x along the inlet flow direction. By continuity $V_1 = Q/A_1$ and $V_2 = Q/A_2$; p_2 follows from Bernoulli's equation (with losses).

Forces on the fluid in the control volume (x-direction): $p_1 A_1 - p_2 A_2 \cos\theta + R_x = \rho Q (V_2 \cos\theta - V_1)$. Force by fluid on the bend is $F_x = -R_x$:

$$F_x = \rho Q (V_1 - V_2 \cos\theta) + p_1 A_1 - p_2 A_2 \cos\theta$$

$$F_y = \rho Q V_2 \sin\theta + p_2 A_2 \sin\theta \text{ (acts opposite to the y-component of outflow)}$$

$$F_R = \sqrt{(F_x^2 + F_y^2)},\ \tan\alpha = F_y / F_x$$

- α = inclination of the resultant force with the x-axis (direction of inlet flow).
- If the bend is in a vertical plane, add the weight of the fluid and bend in the y-direction.
- Large forces on bends of water mains/penstocks need anchor blocks / thrust blocks to resist F_R.
- For a 90° bend of uniform diameter: $V_1 = V_2 = V$, $p_1 \approx p_2 = p$: $F_x = p A$ (and $F_y = \rho Q V + p A$).

Example 6 – Reducing pipe bend

Water flows at 0.1 m^3/s through a horizontal 60° bend reducing from 300 mm to 150 mm. Inlet gauge pressure is 100 kPa. Neglecting losses, find the force on the bend.

Sol: $A_1 = 0.07069\ m^2$; $A_2 = 0.017671\ m^2$; $V_1 = 1.415\ m/s$; $V_2 = 5.659\ m/s$; $\rho Q = 100\ kg/s$

Bernoulli: $p_2 = p_1 + \frac{1}{2}\rho(V_1^2 - V_2^2) = 100000 + 500\ (2.001 - 32.02) = 84990\ Pa = 85.0\ kPa$

$p_{1A1} = 7068.6\ N$; $p_{2A2} = 84990 \times 0.017671 = 1501.9\ N$

$F_x = 100\ (1.415 - 5.659 \times 0.5) + 7068.6 - 1501.9 \times 0.5 = -141.5 + 7068.6 - 751.0 = 6176\ N$

$F_y = (\rho Q V_2 + p_{2A2}) \sin 60° = (565.9 + 1501.9) \times 0.866 = 1791\ N$

$F_R = \sqrt{(6176^2 + 1791^2)} = 6431\ N$; $\alpha = \tan^{-1}(1791/6176) = 16.2°$ to the inlet flow direction.

6. Vortex Flow

A vortex flow is a flow of fluid along curved (concentric circular) paths about an axis. Two types:

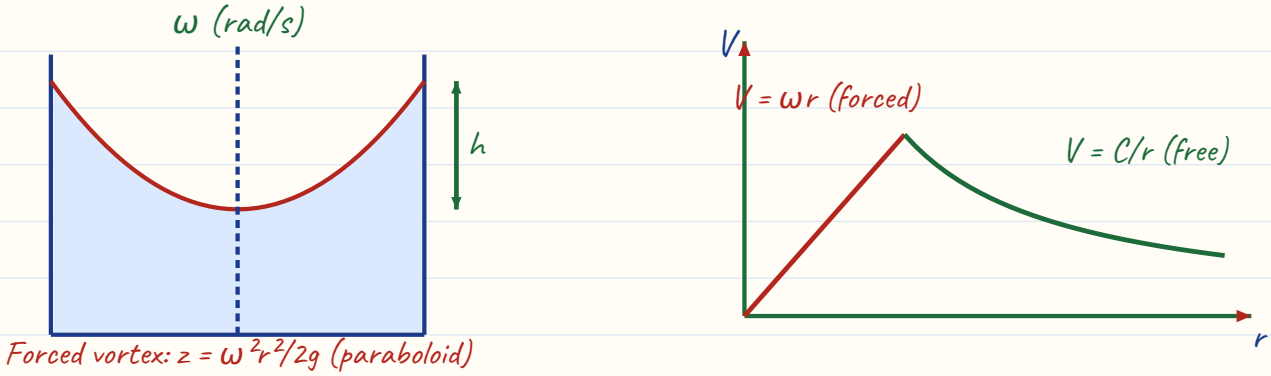


Fig 4.7: Forced vortex (left) and velocity distribution of free and forced vortices (right)

(a) Forced Vortex (rotational, 'solid-body' rotation)

Fluid is rotated by an external torque such that every particle has the same angular velocity ω, e.g. liquid in a cylinder rotated about its vertical axis, centrifugal pump impeller.

- Tangential velocity $V = \omega r$ (increases linearly with r); flow is rotational (vorticity = 2ω).
- Pressure variation: $\partial p/\partial r = \rho \omega^2 r$, $\partial p/\partial z = -\rho g$. Integrating: $p = p_0 + \rho \omega^2 r^2/2 - \rho g z$.
- Free-surface profile ($p = p_{atm}$): $z = \omega^2 r^2 / 2g$ – a paraboloid of revolution measured from the vertex (lowest point on the axis).
- Pressure head at the wall of a closed cylinder of radius R: $h = \omega^2 R^2/2g$ (centrifugal head).
- Volume of paraboloid of revolution = ½ × volume of circumscribing cylinder. Rise of the surface at the wall = fall at the centre = $\omega^2 R^2/4g$ (for a cylinder filled initially to half-way of the paraboloid height, with no uncovering of the bottom).

(b) Free Vortex (irrotational vortex)

Fluid rotates by itself without any external torque, e.g. whirlpool, drain vortex, cyclone, flow around a bend in a river, tornado outside the core.

- Angular momentum is conserved: $V \times r = C$ = constant → $V = C/r$ (velocity is highest near the centre, $V \to \infty$ as $r \to 0$; so a free vortex has a forced-vortex core in practice: Rankine combined vortex).
- Flow is irrotational ($\omega_z = 0$ everywhere except $r = 0$); Bernoulli's equation holds across streamlines, with constant H.
- Pressure: from Bernoulli, $p/\rho g + V^2/2g$ = const → p increases with r ($p_2 - p_1 = \frac{1}{2}\rho (V_1^2 - V_2^2)$) and the free surface profile is $z = z_\infty - C^2/(2 g r^2)$.
- Circulation $\Gamma = 2\pi r V = 2\pi C$ (constant).

Aspect	Forced vortex	Free vortex
Rotation	Rotational ($\zeta = 2\omega$)	Irrotational ($\zeta = 0$)
Velocity	$V = \omega r$ ($\propto r$)	$V = C/r$ ($\propto 1/r$)
Energy	External energy supplied	No external energy, H const
Free surface	$z = \omega^2 r^2/2g$ (paraboloid)	$z = z_\infty - C^2/2gr^2$ (hyperboloid-like)
Examples	Rotating tank, centrifuge, impeller	Whirlpool, drain, cyclone, tornado

Example 7 – Rotating cylinder

An open cylinder of radius 0.15 m and height 0.5 m contains water to a depth of 0.4 m. At what speed can it be rotated about its axis without spilling?

Sol: Spilling begins when the rise at the wall equals 0.5 – 0.4 = 0.1 m. For a paraboloid, rise at wall = fall at centre = $h/2$ where $h = \omega^2 R^2/2g$.

$h = 0.2$ m → $\omega^2 = 2 g h/R^2 = 2 \times 9.81 \times 0.2/0.0225 = 174.4$ → $\omega = 13.2$ rad/s

$N = 60\,\omega/2\pi = 126$ rpm (centre depth = 0.4 – 0.1 = 0.3 m > 0, so the bottom stays covered ✓)

7. Dimensionless Numbers (Dynamic Similarity)

Dimensionless numbers are ratios of two forces acting on a fluid element. For dynamic similarity between a model and prototype, the relevant number must be equal. Let V = velocity, L = characteristic length, ρ = density.

Number	Definition (ratio of forces)	Formula	Significance / Use

Reynolds number Re	Inertia force / Viscous force	$Re = \rho V L / \mu = V L / \nu$	Pipe flow, boundary layers, submarines, flows with viscous forces dominant
Froude number Fr	Inertia force / Gravity force	$Fr = V / \sqrt{(g L)}$	Free-surface flows: open channels, spillways, ships, hydraulic jump; Fr < 1 subcritical, = 1 critical, > 1 supercritical
Mach number M	Inertia force / Elastic (compressibility) force	$M = V / c = V/\sqrt{(K/\rho)}$	Compressible flows: aircraft, rockets, nozzles; M < 1 subsonic, = 1 sonic, > 1 supersonic
Weber number We	Inertia force / Surface-tension force	$We = \rho V^2 L / \sigma$	Capillary waves, droplets, thin films, bubbles, flow over weirs at low heads
Euler number Eu	Pressure force / Inertia force	$Eu = \Delta p/(\rho V^2) = V/\sqrt{(\Delta p/\rho)}$	Pressure-driven flows: cavitation (Eu = cavitation number), turbo-machines, pipe losses

- Reynolds model law: $(\rho V L/\mu)_m = (\rho V L/\mu)_p$ – used for enclosed flows (pipes, valves, flow around submerged bodies).
- Froude model law: $(V/\sqrt{(gL)})_m = (V/\sqrt{(gL)})_p$ – used for free-surface flows (weirs, spillways, ships).
- Speed of sound in a gas: $c = \sqrt{(k R T)}$; in a liquid: $c = \sqrt{(K/\rho)}$.

Example 8 – Dimensionless numbers

Water ($\nu = 1 \times 10^{-6}$ m²/s) flows at 2 m/s in a 50 mm pipe. Find Re. Also find the Froude number for a river flow of V = 3 m/s at depth 1.5 m, and the Mach number of an aircraft at 250 m/s in air at 15 °C.

Sol: $Re = V D/\nu = 2 \times 0.05/10^{-6} = 1 \times 10^5$ (turbulent). $Fr = 3/\sqrt{(9.81 \times 1.5)} = 3/3.836 = 0.78$ (subcritical).

$c = \sqrt{(1.4 \times 287 \times 288)} = 340$ m/s; $M = 250/340 = 0.735$ (subsonic).

Summary of Fluid Dynamics

Concept	Equation
Euler's equation	$dp/\rho + g\,dz + V\,dV = 0$
Bernoulli's equation	$p/\rho g + V^2/2g + z = const$
Venturimeter	$Q = C_d a_1 a_2 \sqrt{(2gh)}/\sqrt{(a_1^2 - a_2^2)}$
Orifice meter	$Q = C_d a_0 a_1 \sqrt{(2gh)}/\sqrt{(a_1^2 - a_0^2)}$
Pitot tube	$V = C_v \sqrt{(2\,g\,h)}$
Momentum equation	$F = \rho\,Q\,(V_2 - V_1)$
Force on bend	$F_x = \rho Q(V_1 - V_{2\cos}\theta) + p_{1A1} - p_{2A2\cos}\theta; F_y = (\rho Q V_2 + p_{2A2})\sin\theta$
Forced vortex	$V = \omega r, z = \omega^2 r^2/2g$
Free vortex	$V\,r = const$

Important Questions

1. Derive Euler's equation of motion and hence Bernoulli's equation. State its assumptions.
2. With a neat sketch, derive the discharge through a venturimeter. Why is its C_d higher than that of an orifice meter?
3. Explain the working of a Pitot tube and derive the velocity expression.
4. State the momentum equation and derive the force exerted by a flowing fluid on a pipe bend.
5. Differentiate between free and forced vortex. Derive the equation of the free surface in a forced vortex.
6. Define Reynolds, Froude, Mach, Weber and Euler numbers with their significance.